Team #1: File System Project – Milestone One

Jiawei Lyu ID: 923809065 Central GitHub Repository: jiaweilyu0929

Leslie Raya ID: 921813630

Alexandra Borders ID: 913630008

Yeraldin Crespo ID: 923523819

Repository Link: https://github.com/jiaweilyu0929/csc415-group-project.git

1. Dump from HexDump utility

```
student@student: ~/CSC415/csc415-group-project — ./fsshell SampleVol...
~/CSC415/csc415-group-project

student@student:~/CSC415/csc415-group-project$ rm SampleVolume
./fsshell SampleVolume 9999872 512
File SampleVolume does not exist, errno = 2
File SampleVolume not good to go, errno = 2
Block size is : 512
Created a volume with 9999872 bytes, broken into 19531 blocks of 512 bytes.
Opened SampleVolume, Volume Size: 9999872;  BlockSize: 512; Return 0
Initializing File System with 19531 blocks with a block size of 512
No valid filesystem found; formatting...
Volume formatted: 19531 blocks, 512 bytes/block, data starts at LBA 10
File system ready.
|---------------------------------|
|------- Command ------|- Status -|
| ls                  |    OFF   |
| cd                  |    OFF   |
| md                  |    OFF   |
| pwd                 |    OFF   |
| touch               |    OFF   |
| cat                 |    OFF   |
| rm                  |    OFF   |
| cp                  |    OFF   |
| mv                  |    OFF   |
| cp2fs               |    OFF   |
| cp2l                |    OFF   |
|---------------------------------|
Prompt >
```

```
System exiting
student@student:~/CSC415/csc415-group-project$ ./Hexdump/hexdumpl SampleVolume --start 1 --count 1
Dumping file SampleVolume, starting at block 1 for 1 block:

000200: 31 43 53 46 01 00 00 00  00 02 00 00 00 00 00 00 | 1CSF************
000210: 4B 4C 00 00 00 00 00 00  41 4C 00 00 00 00 00 00 | KL******AL******
000220: 01 00 00 00 00 00 00 00  05 00 00 00 00 00 00 00 | ****************
000230: 06 00 00 00 00 00 00 00  04 00 00 00 00 00 00 00 | ****************
000240: 0A 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000250: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000260: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000270: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000280: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000290: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000300: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000310: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000320: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000330: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000340: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000350: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000360: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000370: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000380: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000390: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

student@student:~/CSC415/csc415-group-project$
```

```
student@student:~/CSC415/csc415-group-project$ ./Hexdump/hexdumpl SampleVolume --start 2 --count 5
Dumping file SampleVolume, starting at block 2 for 5 blocks:

000400: FF 03 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ?****************
000410: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000420: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000430: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000440: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000450: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000460: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000470: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000480: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000490: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000500: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000510: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000520: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000530: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000540: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000550: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000560: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000570: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000580: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000590: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
000600: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000610: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000620: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000630: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000640: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000650: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000660: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000670: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000680: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000690: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000700: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000710: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000720: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000730: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000740: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000750: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000760: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000770: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000780: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000790: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000800: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000810: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000820: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000830: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
000830: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000840: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000850: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000860: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000870: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000880: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000890: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0008A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0008B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0008C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0008D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0008E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0008F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000900: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000910: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000920: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000930: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000940: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000950: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000960: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000970: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000980: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000990: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0009A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0009B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0009C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0009D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0009E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0009F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000A00: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
000A60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A80: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000A90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000AA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000AB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000AC0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000AD0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000AE0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000AF0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000B00: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B80: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000B90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000BA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000BB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000BC0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000BD0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000BE0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000BF0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000C00: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C80: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
000C90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CC0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CD0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CE0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CF0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000D00: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D80: 00 00 00 00 00 00 00 00  00 F8 FF FF FF FF FF FF | *********???????
000D90: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DA0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DB0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DC0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DD0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DE0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DF0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????

student@student:~/CSC415/csc415-group-project$
```

```
student@student:~/CSC415/csc415-group-project$ ./Hexdump/hexdumpl SampleVolume --start 6 --count 4
Dumping file SampleVolume, starting at block 6 for 4 blocks:

000C00: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C80: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000C90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CC0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CD0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CE0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000CF0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000D00: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000D80: 00 00 00 00 00 00 00 00  00 F8 FF FF FF FF FF FF | *********???????
000D90: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DA0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DB0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DC0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DD0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DE0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????
000DF0: FF FF FF FF FF FF FF FF  FF FF FF FF FF FF FF FF | ????????????????

000E00: 2E 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | .***************
000E10: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E20: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E30: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E80: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000E90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000EA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
000E90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000EA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000EB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000EC0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000ED0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000EE0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000EF0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000F00: 02 01 06 00 00 00 00 00  00 00 04 00 00 00 00 00 | ****************
000F10: 00 00 00 08 00 00 00 00  00 00 BC 21 D7 69 00 00 | **********?!?i**
000F20: 00 00 BC 21 D7 69 00 00  00 00 BC 21 D7 69 00 00 | **?!?i****?!?i**
000F30: 00 00 2E 2E 00 00 00 00  00 00 00 00 00 00 00 00 | **..************
000F40: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000F50: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000F60: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000F70: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000F80: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000F90: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000FA0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000FB0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000FC0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000FD0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000FE0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000FF0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

001000: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001010: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001020: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001030: 00 00 02 01 06 00 00 00  00 00 00 00 04 00 00 00 | ****************
001040: 00 00 00 00 00 08 00 00  00 00 00 00 BC 21 D7 69 | ************?!?i
001050: 00 00 00 00 BC 21 D7 69  00 00 00 00 BC 21 D7 69 | ****?!?i****?!?i
001060: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001070: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001080: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001090: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0010A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0010B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0010C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0010D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0010E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0010F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

001100: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001110: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001120: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001130: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001140: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001150: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
001130: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001140: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001150: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001160: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001170: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001180: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001190: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0011A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0011B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0011C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0011D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0011E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0011F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

001200: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001210: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001220: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001230: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001240: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001250: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001260: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001270: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001280: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001290: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0012A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0012B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0012C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0012D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0012E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0012F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

001300: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001310: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001320: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001330: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001340: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001350: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001360: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001370: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001380: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
001390: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0013A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0013B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0013C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0013D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0013E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0013F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

student@student:~/CSC415/csc415-group-project$
```

2. Volume Control Block Description

The Volume Control Block is the first structure written to the volume and is stored at block 0. This serves as the entry point for the entire filesystem. Each time the system initializes, it reads the VCB first to then locate all of the other components.

The *magic* field (0x46534331) allows the system to determine at startup whether the volume has already been formatted. When *fsInit* runs, it reads block 0 and checks this value. If it is a match, then the volume is valid and the remaining fields can be trusted. If it does not match, the system treats the volume as unformatted and initializes everything from scratch.

The *block_size* and *total_blocks* fields define the geometry of the volume. All other locations stored in the VCB are expressed as block numbers rather than byte offsets, so these two values are fundamental to navigating the disk.

The *bitmap_start_lba* and *bitmap_block_count* fields tell the system where the bitmap lives. Without these functions, the filesystem would have no way to locate free space after a restart. Similarly, *root_dir_start_lba* and *root_dir_block_count* allow the system to find the root directory without searching the entire volume.

The *free_block_count* field tracks how many blocks are currently available, and *first_data_lba* marks where the actual file data begins.

3. Free Space Structure Description

We use a bitmap to track free space on the volume – which blocks on the volume are available and which are already in use. Each block on the disk is represented by one bit in the bitmap. If the bit is 0, the block is free. If the bit is 1, the block is already in use. This approach is memory efficient because we only need one bit per block rather than storing a full list of free blocks.

The bitmap is stored starting at block 1, immediately after the VCB. Its size depends on the total number of blocks in the volume. For our default volume of 19,531 blocks, we need 19,531 bits, which equals about 2,442 bytes. Using the ceiling formula *(bytes + block_size - 1) / block_size*, this rounds up to 5 blocks of storage for the bitmap.

When the volume is first formatted, *initFreeSpace* sets up the bitmap in memory using *calloc*, which zeros out all the bits marking every block as free. It then marks the blacks already in use – block 0 for the VCB, blocks 1 through 5 for the bitmap itself, and blocks 6 through 9 for the root directory – as used. Any bits beyond the total block count are also marked as used to prevent invalid allocations. The completed bitmap is then written to disk with *LBAwrite*.

The bitmap is kept in memory as a global pointer *g_bitmap* while the system is running. This means we never need to re-read it from the disk during normal operation – we simply update the in-memory copy and write it back to disk any time a block is allocated.

The *allocateBlocks* function handles allocation by scanning the bitmap for a contiguous run of free blocks. It tracks consecutive free bits using a counter, and once enough are found, it marks them all as used, decreases the free block count in the superblock, and flushes the updated bitmap to disk so the change is permanently saved.

4. Directory System Description

Directories are stored on disk as arrays of fixed-size *fs_dirent_t* entries, where the total number of entries is determined by dividing the total byte size of the directory by the size of a single entry. Each entry contains the following fields: name for the file or directory name, *fileType* to distinguish between files and directories, *inUse* to indicate whether the slot is occupied, *startBlock* for the first block of the entry's data, *blockCount* for how many blocks it occupies, and size for the total byte size. Three timestamp fields are also included: createTime, modifiedTime, and accessTime, used for tracking the history of the entry. The root directory is allocated a fixed number of blocks defined by *FS_ROOT_DIR_BLOCKS* at format time, and the entire directory is written to disk in one call to *LBAwrite.*

When the root directory is initialized by *initRootDir*, the first two slots are always reserved for the dot and dot dot entries. The dot entry has its startBlock and blockCount set to point back to the root directory itself, representing the current directory. The dot dot entry is set identically since the root directory has no parent, so both entries point to the same location on disk. All remaining slots in the directory are left as zeroed memory from the initial *calloc*, meaning that the inUse field is 0, and they are treated as empty and available for future file and directory creation.

5. Table of who worked on which components

Component	**Team Member**
VCB, Directory	Jiawei Lyu
Directory	Leslie Raya
VCB	Alexandra Borders
Free Space	Yeraldin Crespo

6. Team Work

Our team worked together by staying in communication and helping each other throughout the project. We met a few times during the week to check in on progress and make sure we were all on the same page. Most of our meetings were through Zoom and Discord, since it made it easy to share our screens and go over code together. We split up the work based on the main parts of the assignment. Each person focused on one area, like the VCB, free space, or root directory, but we still helped each other when someone got stuck or had questions. This made it easier to move faster while still understanding how everything connects.

We used GitHub to keep all our code in one place and to combine our work. Everyone pushed their changes regularly, and we checked things together to make sure nothing was breaking. When we ran into issues, we either talked it through during meetings or messaged each other to figure it out. Overall, we worked in a pretty organized way, balancing working independently and helping each other when needed, which helped us complete the milestone successfully.

7. Issues & Resolutions

One of the early challenges our team faced was getting everyone set up with the shared repository. Since the project started from the main file system project, we had to make sure everyone cloned the correct team repo, had the right files, and were properly synced before starting to code.

Each of us had our own confusion during the process, but some of those issues included:

- Confusion when adding the *fs_dirent_t* struct, where the fields were accidentally merged into the existing *fs_superblock_t* struct instead of being defined as a separate structure. We resolved this by using the undo feature in VS Code to revert the changes and then carefully adding the new struct separately below the existing one.
- Another issue was function ordering in C – when *initFileSystem* tried to call *fs_vol_format* before it was defined, the compiler threw errors. We resolved with forward declarations at the top of the file.
- Finally, we had an issue where the *SampleVolume* file persisted between tests, causing the system to mount the old volume instead of formatting a fresh one. We learned to always delete the volume file with *rm SampleVolume* before testing changes.